Monte Carlo DCF Enterprise Value Analysis

Oshkosh Corporation (OSK) | 50,000 simulations | Five-year forecast

All monetary values are US$ millions unless otherwise stated.

1. Executive Summary

This project applies Monte Carlo simulation to a discounted cash flow framework to estimate a probability distribution for the enterprise value of Oshkosh Corporation rather than relying on a single deterministic valuation. The simulation varies key operating and valuation assumptions across 50,000 trials and compares the resulting model-implied enterprise values with the current enterprise value of $9,060 million.

Measure	Model result	Interpretation
Mean model-implied EV	$6,673.32m	26.3% below current EV
Median model-implied EV	$6,663.82m	26.4% below current EV
5th percentile EV	$3,422.73m	62.2% below current EV
95th percentile EV	$9,930.34m	9.6% above current EV
Probability EV > $9,060m	11.234%	Low probability of model-implied upside
Probability EV ≤ $9,060m	88.766%	High probability of overvaluation

Overall conclusion. The model produces a median enterprise value of $6,663.82 million, materially below the current enterprise value of $9,060 million. Only 11.234% of simulations exceed the current EV, while the 5th-to-95th percentile range is approximately $3.42 billion to $9.93 billion. Under the simulated assumptions, OSK appears overvalued and offers a limited margin of safety at the current enterprise value.

2. Project Objective

The objective is to estimate a defensible range of possible enterprise values for OSK by combining a five-year unlevered DCF with Monte Carlo simulation. The approach recognises that sales growth, profitability, capital expenditure, tax, working capital, the weighted average cost of capital, and perpetual growth are uncertain rather than fixed.

- Produce a valuation distribution instead of one point estimate.
- Measure downside, central, and upside valuation outcomes using percentiles.
- Estimate the probability that model-implied enterprise value exceeds the current enterprise value.
- Identify the assumptions that have the strongest relationship with enterprise value.

3. Historical Basis and Model Inputs

The simulation inputs were calibrated primarily from the historical financial analysis in the accompanying workbook. OSK reported 2024 sales of $10,755.5 million, operating income of $1,010.7 million, depreciation and amortisation of $200.1 million, capital expenditure of $232.0 million, and operating net working capital of $1,786.5 million. Historical analysis also showed a 2024 EBIT margin of 9.397%, capex equal to 2.157% of sales, and operating working capital equal to 16.610% of sales. These figures provide the starting point and ranges used in the simulation.

Input	Distribution / range	Economic role	Implementation
Sales growth	Year-specific triangular	Drives forecast revenue	Independently simulated

	ranges; modes decline from 12% to 7%		by year
EBIT margin	Triangular: 4.495% / 7.458% / 9.397%	Determines operating profit	One draw held constant across five years
WACC	Triangular: 8% / 9% / 11%	Discounts FCFF and terminal value	One draw held constant across five years
Perpetual growth	Triangular: 1.5% / 2.5% / 3.1%	Drives terminal value	One draw per simulation
Capex / sales	Triangular: 2.157% / 3.000% / 3.913%	Cash reinvestment requirement	One draw held constant across five years
Tax rate	Triangular: 21.0% / 23.3% / 26.0%	Converts EBIT to NOPAT	One draw held constant across five years
Operating NWC / sales	Triangular: 16.6% / 16.7% / 17.7%	Working-capital investment	Converted to annual NWC balances; year-on-year changes deducted from FCFF
D&A	$200.1m grown with cumulative sales	Non-cash add-back	Scales with revenue growth

4. Methodology

For each of 50,000 trials, the model generates a five-year financial path and calculates enterprise value using the following sequence:

- Forecast sales by compounding the simulated annual sales-growth rates from the 2024 revenue base.
- Estimate EBIT as forecast sales multiplied by the simulated EBIT margin.
- Calculate NOPAT as EBIT × (1 − tax rate).
- Estimate FCFF as NOPAT + depreciation and amortisation − capital expenditure − change in net working capital.
- Discount each annual FCFF by the simulated WACC.
- Calculate terminal value using the Gordon growth model: Terminal FCFF ÷ (WACC − perpetual growth).
- Add the present value of forecast-period FCFF and terminal value to obtain model-implied enterprise value.

Core equation: Enterprise Value = $\Sigma[FCFF_t \div (1 + WACC)^t] + [\text{Terminal Value} \div (1 + WACC)^5]$.

5. Simulation Results

The simulated distribution is approximately bell-shaped around the central valuation, with a modest right tail. The median of $6,663.82 million is slightly lower than the mean of $6,673.32 million. The histogram marks the 5th, 50th, and 95th percentiles at $3,422.73 million, $6,663.82 million, and $9,930.34 million, respectively.

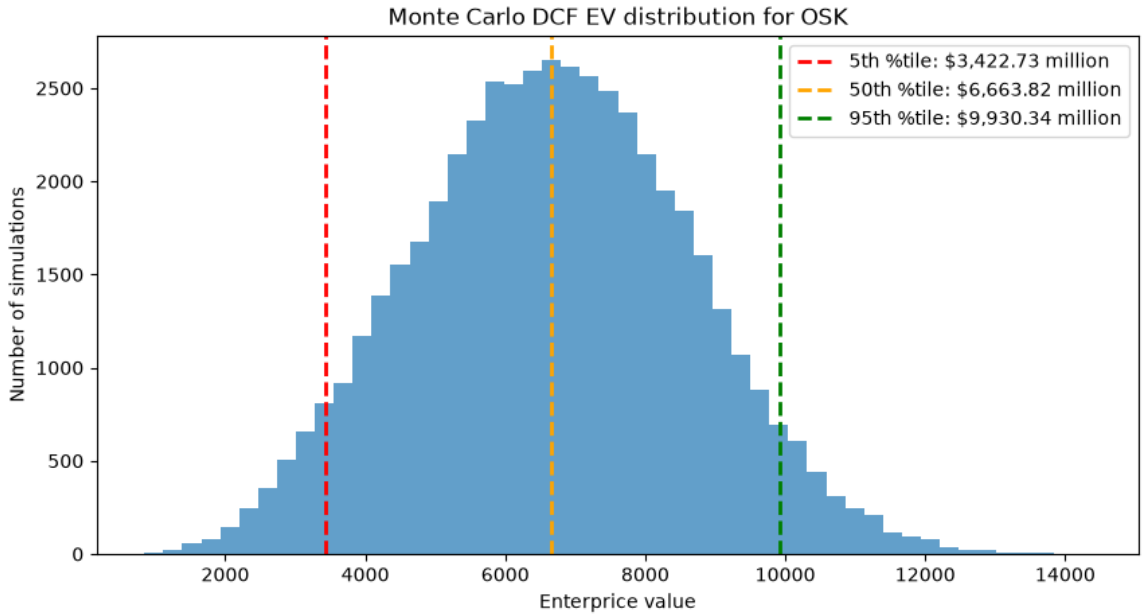


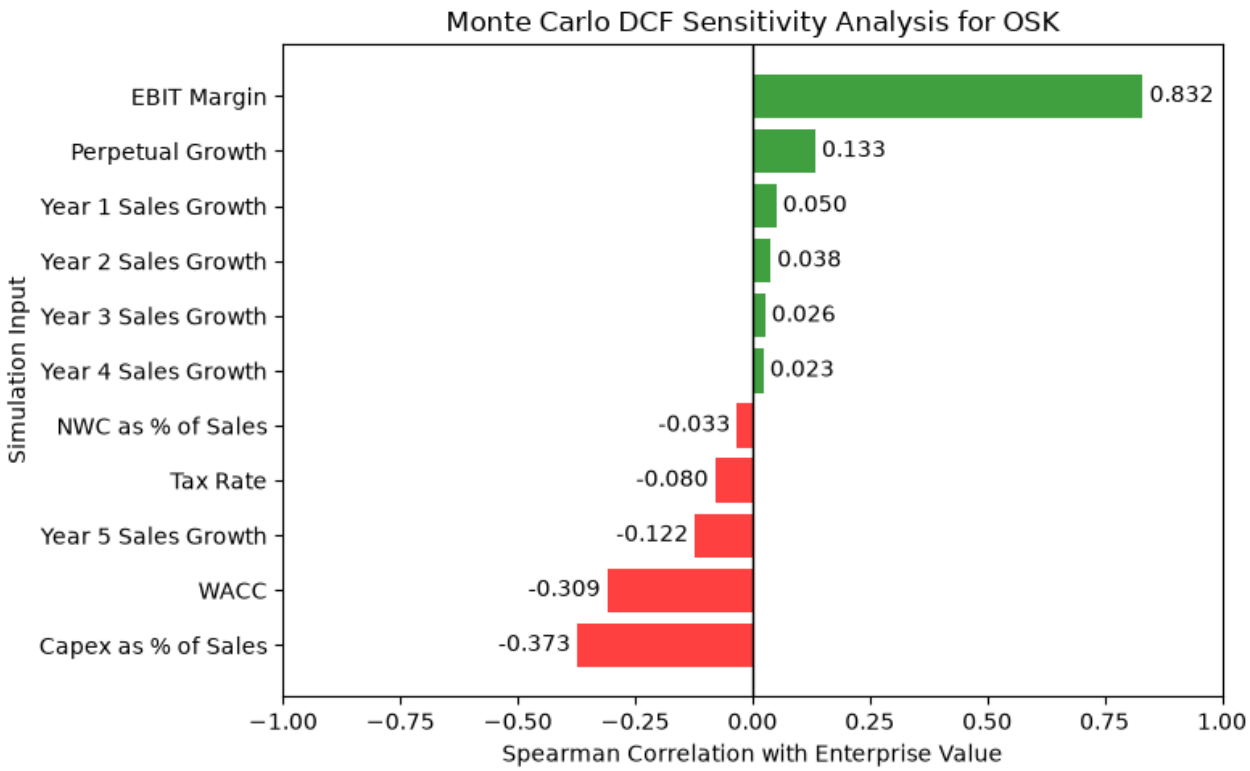
Figure 1. Monte Carlo DCF enterprise-value distribution for OSK. The red, orange, and green dashed lines represent the 5th, 50th, and 95th percentiles.

Scenario interpretation

Scenario definition	Enterprise-value range	Interpretation
Lower third / conservative	Below $5,792.07m	Materially below current EV; adverse operating or valuation outcomes
Middle third / base	$5,792.07m–$7,542.50m	Central range remains below the current EV of $9,060m
Upper third / optimistic	Above $7,542.50m	More favourable outcomes, but only 11.234% exceed current EV
90% probability interval	$3,422.73m–$9,930.34m	Broad uncertainty range between 5th and 95th percentiles

Market comparison. The current enterprise value of $9,060 million is $2,386.68 million above the simulation mean and $2,396.18 million above the median. The probability of model-implied enterprise value exceeding the current value is 11.234%, while the probability of being equal to or below the current value is 88.766%. The model therefore indicates an unfavourable valuation balance at the current EV.

6. Sensitivity Analysis



The sensitivity analysis shows that EBIT margin is the dominant valuation driver, with a strong positive Spearman correlation of 0.832 with enterprise value. Capital expenditure as a percentage of sales and WACC are the largest negative drivers, with correlations of -0.373 and -0.309, respectively. Perpetual growth has a moderate positive relationship of 0.133. The negative Year 5 sales-growth correlation of -0.122 reflects higher terminal-year working-capital investment under the current model and should be interpreted cautiously rather than as evidence that growth destroys value.

7. Investment Interpretation

- Central valuation: The mean and median enterprise values are approximately $6.67 billion, materially below the current EV of $9.06 billion.
- Downside risk: The 5th-percentile EV of $3.42 billion is approximately 62.2% below the current EV, indicating substantial downside under adverse combinations of operating and valuation assumptions.
- Upside potential: The 95th-percentile EV of $9.93 billion is approximately 9.6% above the current EV, showing that only favourable combinations of assumptions provide meaningful upside.
- Decision quality: Only 11.234% of simulations exceed the current EV, so the model does not support a standalone undervaluation or buy signal.
- Key value drivers: EBIT margin is the strongest overall driver. Capex intensity and WACC are the principal negative drivers, highlighting the importance of operating profitability, reinvestment requirements and discount-rate selection.
- Use in practice: The output is most useful as a valuation-risk framework for investment screening, scenario planning and identifying assumptions that require further due diligence.

8. Conclusion

The Monte Carlo DCF produces a mean enterprise value of $6.67 billion and a median of $6.66 billion, both materially below the current EV of $9.06 billion. The 11.234% probability of exceeding current EV indicates

limited upside under the modelled assumptions, while the 5th-to-95th percentile range of $3.42 billion to $9.93 billion demonstrates substantial uncertainty. The preliminary investment conclusion is therefore cautious to negative: OSK does not provide a clear margin of safety at the current enterprise value. EBIT margin is the most important positive driver, while capex intensity and WACC are the main negative drivers.

Appendix: Working-Capital Validation

The model forecasts operating NWC as a percentage of sales and deducts only the annual increase in the NWC balance from FCFF:

$$\textit{Forecast } NWC_t = \textit{Forecast } Sales_t \times \textit{NWC-to-Sales Ratio}$$
$$\textit{Change in } NWC_1 = \textit{Forecast } NWC_1 - \textit{Initial NWC}$$
$$\textit{Change in } NWC_t = \textit{Forecast } NWC_t - \textit{Forecast } NWC_{t-1}$$